

LFM2.5-Audio on AI Glasses: Component-Level Feasibility on a Qualcomm NPU Proxy

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Abstract

We evaluate whether LFM2.5-Audio-1.5B is deployable as a speech-to-speech assistant on AI glasses. Because no AR1-class target was available, we profile fixed-shape component exports on a Qualcomm QCS8550 proxy, establish a full-model BF16 reference on an NVIDIA L4, and run a local Q4 quantization matrix on Apple M2 Metal. Every tested neural partition compiles to a strict NPU context with zero CPU fallback, and the model sustains two-turn interleaved text and audio generation on the reference GPU. Placement, however, is not acceptance: the FP16 detokenizer places all 371 runtime layers on the NPU and still returns unusable audio (waveform cosine 0.0052, SI-SDR -45.74 dB) on real generated codes. We therefore continue LFM2.5-Audio as a gated research baseline rather than a glasses-ready result, and profile a sub-1B modular pipeline (Moonshine Base $\rightarrow$ Qwen3-0.6B $\rightarrow$ Pocket TTS) as deployment control. We report the component evidence, its boundaries, and the acceptance gates that remain open.

1 Introduction

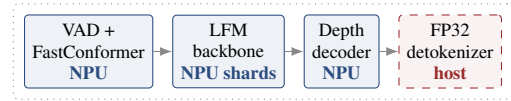
Speech-to-speech assistants on eyewear are constrained less by model quality than by what a wearable NPU will accept and what its RAM budget will hold. This report asks a narrow question about one candidate: can LFM2.5-Audio-1.5B [Liquid AI, 2026a] serve as the integrated baseline for a first device integration, and what evidence would be needed to call it glasses-ready?

The answer is a qualified yes on the research track and a clear no on the product track. All fixed-shape neural partitions we tested compiled to strict NPU contexts with zero CPU fallback on the proxy device, and the official BF16 model completed ASR and a persistent two-turn speech interaction on an NVIDIA L4. The blockers are target-hardware verification, target-QNN resident memory, a production cache interface, and the output-audio path.

Hardware boundary. Every Qualcomm number in this report was measured on QCS8550 (Proxy), Android 12, Hexagon v73, through Qualcomm AI Hub [Qualcomm, 2026]. No AR1/AR1+/5100 target was available. Proxy figures are *placement evidence*, not AR1 performance, and this qualification applies to every Qualcomm table and figure below without further repetition.

Contributions.

- A strict-placement component profile of LFM2.5-Audio on a Hexagon NPU, with per-component latency, peak memory, and numerical acceptance (§3).
- Evidence that full NPU placement and numerical usability are independent properties, demonstrated by a fully-placed detokenizer that fails on real codes (§5).



Session manager: bounded context, reset/summary policy

Figure 1: Recommended LFM partition for a first device integration. The host detokenizer is an architectural recommendation, not a measured on-glasses result.

- A static memory and KV-cache ledger showing why BF16 assets cannot be resident in a 2 GiB budget (§6).
- A candidate comparison against Whisper, Moonshine, Zipformer, and Piper component cards, and the resulting baseline decision (§8).

2 Setup

2.1 Usage mode

The product mode is VAD-gated activation followed by a continuous conversational session. This avoids full-model inference while idle, but it does not bound active-session context: once a session opens, KV growth is unbounded until the session manager intervenes. An explicit reset, summarization, or eviction policy is therefore a requirement, not a refinement. Figure 1 shows the partition we recommend for a first integration.

2.2 Evidence classes

Table 1 states what each class of measurement does and does not establish. The distinction matters throughout: local, proxy, and reference-GPU evidence are not interchangeable, and none of them is a glasses measurement.

Table 1: Scope of the experiments. The final row is the part that constrains every conclusion in this report.

Class	Contents
Local	Eight official-weight fixed-shape PT2 exports; repeated Q4_0/KV-cache matrix on Apple M2 Metal; static memory ledger.
Qualcomm proxy	Strict NPU placement, component-only latency and memory, local-vs-remote numerical comparison.
Reference GPU	Full BF16 ASR plus two-turn interleaved text/audio generation on one NVIDIA L4.
Quality	Downstream encoder sensitivity, exact depth tokens, generated-wave integrity, small matched ASR diagnostic.
Not measured	Full AR1 pipeline, host/NPU transfers, real glasses RSS, power, thermals, VAD, microphone-to-speaker latency.

Table 2: Strict-placement component results on QCS8550 (Proxy). NPU/CPU is the operator split; every graph placed with zero CPU fallback. Numerics: ✓ pass, ✗ mismatch against the local golden.

Component	NPU/CPU	Latency	Peak	Num.
FastConformer + adapter	1215/0	4.786 ms	87.0 MiB	✗
Conv layer prefill	36/0	2.693 ms	92.2 MiB	✓
Attention layer prefill	62/0	2.444 ms	91.5 MiB	✓
Conv cached decode	32/0	2.695 ms	91.9 MiB	✓
Attention cached decode	61/0	2.452 ms	87.0 MiB	✓
Depth/RQ decoder	2953/0	25.540 ms	92.0 MiB	✓
Detokenizer neural $T=4$	371/0	1.880 ms	88.0 MiB	✗
Detokenizer neural $T=8$	371/0	2.577 ms	87.4 MiB	✗

3 Component placement on the proxy

Every tested strict graph placed fully on the Hexagon NPU (Table 2). Zero operators fell back to CPU in any partition. Latency spans an order of magnitude, from 1.880 ms for the four-frame detokenizer to 25.540 ms for the depth/RQ decoder, and peak memory is flat at 87–92 MiB across all eight graphs.

The two ✗ rows are the finding. Placement and numerical acceptance are independent: the detokenizer compiles cleanly, runs fast, and produces audio that is unusable (§5).

4 Full-model reference behavior

The official BF16 model completed a persistent two-turn speech interaction on a single NVIDIA L4 (Table 3). Turn 2 is uniformly faster than turn 1 because the session cache is warm: first text drops from 97.97 ms to 56.49 ms and first audio token from 339.67 ms to 225.87 ms.

The run used 3.034 GB steady and 6.216 GB peak CUDA allocation. These are reference-GPU allocator figures, not mobile RSS, and they are not a budget for the glasses. The official output detokenizer remained FP32 throughout.

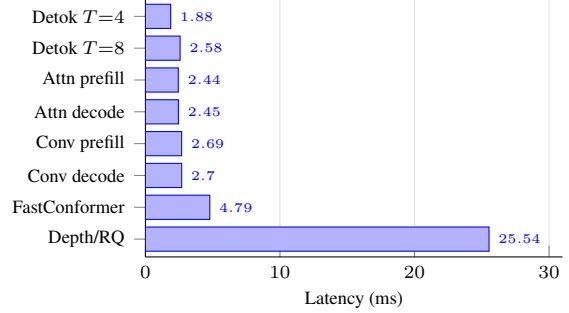


Figure 2: Per-component latency on the proxy. These values must not be summed: the components have different shapes, different invocation counts, different compiled-context overheads, and different transfer boundaries.

Table 3: Full-model BF16 golden on one NVIDIA L4. Turn 2 benefits from a warm session cache.

Metric	Turn 1	Turn 2
First text	97.97 ms	56.49 ms
First audio token	339.67 ms	225.87 ms
Generated steps/s	15.74	16.35
Decoded waveform	11.84 s	8.96 s

5 Quality

5.1 Encoder fidelity

Proxy FastConformer features reach cosine 0.998424 and NRMSE 0.06976 against the local golden, with 100% nearest-frame temporal identity over the ten captured frames. A frozen downstream LFM sensitivity test preserves 16/17 top-1 decisions; the single branch change is *seems to looks*, and the golden choice remains in the remote top-five at every step.

The encoder path is therefore good enough to carry the baseline. But exact tensor closeness is not the gate. The detokenizer below has excellent placement and catastrophic output, which is precisely why every quantization change must be re-checked on transcripts, tokens, and task-level behavior rather than on tensor distance alone.

5.2 Detokenizer failure

The depth/RQ decoder returns exact audio code tokens. The FP16 NPU detokenizer, given those real generated codes, does not: waveform cosine 0.0052, NRMSE 1.010, and SI-SDR −45.74 dB over the first eight valid turn-1 frames. The graph still places 371/371 runtime layers on the NPU at 2.564 ms.

This is the single most consequential result in the report. A component can be fully accepted by the compiler, run at target speed, and be worthless. The first device integration should keep the official FP32 detokenizer outside the FP16 HTP path until a host implementation is measured and accepted.

Table 4: WER (%) on the same 18-utterance diagnostic; lower is better. Best per column in bold.

Model	Clean	White 10 dB	Compet. 5 dB
LFM2.5	2.86	6.43	55.71
Whisper Tiny	11.43	21.67	71.90
Whisper Base	8.81	20.71	55.00
Whisper Small	7.14	10.71	31.43
Moonshine Base	8.57	14.52	64.05

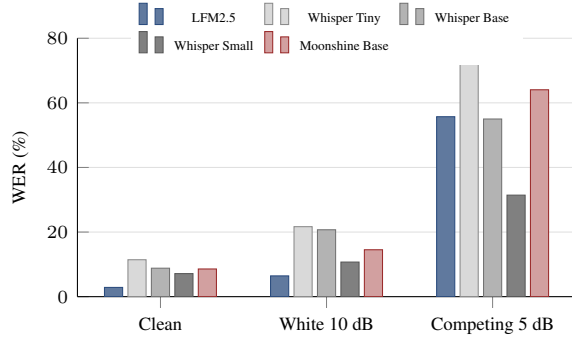


Figure 3: WER on the 18-utterance diagnostic. The clean-speech ranking does not survive competing speech.

5.3 Matched ASR diagnostic

We transcribed the same 18 LibriSpeech dummy utterances under three conditions: clean, deterministic 10 dB white noise, and 5 dB competing speech (Table 4, Figure 3). WER is normalized; lower is better. This is a smoke diagnostic on synthetic noise, not a product benchmark and not a glasses-microphone benchmark.

LFM is strongest on clean speech (2.86%) and on 10 dB white noise (6.43%). Under 5 dB competing speech the ordering inverts: Whisper Small leads at 31.43% while LFM degrades to 55.71%. Moonshine Base remains the footprint challenger, not a demonstrated multi-speaker quality winner.

5.4 Generated-audio intelligibility

Retranscribing LFM’s own generated audio with Whisper Small gives 0.00% WER for turn 1 and 21.43% WER for turn 2 against LFM’s own generated text. The WAVs are finite, 24 kHz mono, and unclipped. This is an audio-text consistency proxy, not a naturalness or MOS score.

6 Memory and context budget

The pinned BF16 runtime checkpoint set occupies 3.390 GiB in static files. The complete local Q4 GGUF bundle occupies 1.001 GiB — a 3.39× smaller package, though across different formats and backends, so it is a package-to-package comparison and nothing stronger. Static file size is not process RSS. The consequence stands regardless: the BF16 assets cannot all be resident inside a 2 GiB RAM budget.

Table 5: Ideal KV payload by context length. Tensor-payload estimates only; actual quantized runtime buffers differ because of scales and alignment.

Context positions	FP16 payload	INT8 payload
4,096	48 MiB	24 MiB
8,192	96 MiB	48 MiB
16,384	192 MiB	96 MiB
32,768	384 MiB	192 MiB

Active-session growth is dominated by the six attention-layer KV caches (Table 5); the fixed convolution cache is roughly 0.12 MiB at BF16 and never matters. VAD reduces idle work but does not touch this requirement — which is why the session policy of §2.1 is load-bearing.

7 Local quantization matrix

The official Q4_0 GGUF bundle was run 40 times across two official audio samples with FP16, Q8_0, and Q4_0 KV caches plus a flash-attention control. All 40/40 normalized transcripts match exactly (Table 6). Q4_0 KV cuts cache memory from 48.0 MiB to 13.5 MiB with no transcript change on this sample.

This is an Apple M2 Metal sanity check. It is not Qualcomm QNN evidence, not AR1 evidence, and not a general quality benchmark. The runner reports Metal fallback hotspots for CONV_2D_DW, ROLL, and UNARY in the audio encoder; those are Metal hotspots, not QNN CPU-fallback evidence. No quantized LFM QNN graph has yet been profiled on QCS8550 or on the glasses.

8 Comparators and candidate selection

8.1 Qualcomm speech component cards

Table 7 reports full-NPU proxy scorecards from qai-hub-models 0.57.3 for the alternative pipeline components. These are fixed component invocations, not end-to-end pipeline latency.

Whisper Small W8A16 is instructive: quantization makes the encoder 2.89× slower while making the decoder roughly 35% faster. Quantization must therefore be profiled stage by stage; a single whole-model figure would have hidden both effects.

8.2 Decision

Table 8 records the candidate outcome. LFM2.5-Audio continues as the integrated research baseline under the gates of §9. Moonshine Base → Qwen3-0.6B → Pocket TTS [Useful Sensors, 2024, Qwen Team, 2025, Kyutai Labs, 2025] is the recommended next challenger: it creates the most weight-footprint headroom while keeping the text model on a known Qualcomm path. It still needs export, quality, and host-TTS measurements before it can replace LFM.

LLaMA-Omni2-0.5B [Fang et al., 2025] is excluded on

Table 6: Local Q4 inference-technique matrix on Apple M2 Metal. Values are median/p95 over five runs per sample and configuration. CLI model time excludes model load; wall time in the public data includes process startup and file-cache effects.

KV	Flash	Sample	Exact	KV MiB	Encode ms <i>med/p95</i>	Gen tok/s <i>med/p95</i>	Model ms <i>med/p95</i>
FP16	off	asr	5/5	48.0	449.0 / 629.8	24.3 / 25.6	3425.8 / 3555.5
FP16	off	question	5/5	48.0	140.0 / 298.4	29.1 / 57.4	816.9 / 1590.3
FP16	on	asr	5/5	48.0	439.0 / 592.4	23.2 / 26.4	3389.3 / 3855.5
FP16	on	question	5/5	48.0	172.0 / 196.8	28.6 / 38.8	891.4 / 1096.6
Q4_0	on	asr	5/5	13.5	413.0 / 476.8	25.7 / 26.1	3122.0 / 3285.7
Q4_0	on	question	5/5	13.5	139.0 / 148.8	32.6 / 48.6	738.1 / 904.6
Q8_0	on	asr	5/5	25.5	486.0 / 780.4	26.5 / 28.1	3254.8 / 3780.5
Q8_0	on	question	5/5	25.5	137.0 / 195.4	41.7 / 67.7	627.5 / 982.5

Table 7: Qualcomm proxy component cards for comparator models (`qai-hub-models` 0.57.3). Fixed component invocations, not end-to-end pipeline latency.

Model	Encoder/main	Recurrent	Use
Zipformer	8.895 ms / 0.71 s chunk	0.075 ms dec.; 0.187 ms joiner	Streaming fallback
Whisper Tiny	25.307 ms	2.459 ms dec.	Small control
Whisper Base	47.978 ms	4.202 ms dec.	Middle control
Whisper Small FP16	130.318 ms	12.074 ms dec.	Overlap-robust control
Whisper Small W8A16	376.895 ms	7.856 ms dec.	Memory tradeoff
Piper TTS	30.344 ms enc.; 15.189 ms flow	3.018 ms dec.	Audio fallback

Table 8: Candidate decisions. Footprint figures come from different sources and are not directly comparable.

Candidate	Footprint view	Decision
LFM2.5-Audio	Q4 bundle 1.001 GiB locally	Continue with gates
Moonshine	+ Est. 0.55–0.66 GB	Profile next
Qwen3-0.6B	+ weights	
Pocket		
Zipformer	+ ~0.84 GB from compo-	Demo insurance
Qwen3-0.6B	+ nent cards	
Piper		
Mini-Omni [Xie & Wu, 2024]	~976M total est.	Later control
LLaMA-Omni2-0.5B	Checkpoint alone 3.857 GB BF16	Exclude

evidence rather than principle: the size label refers to the language backbone, while the checkpoint alone is 3.857 GB in BF16.

9 Gates and next steps

9.1 Demo acceptance gates

Table 9 lists provisional thresholds for a demo. They are provisional in a specific sense: the memory gate in particular is a guess until app-available RAM on the real device is confirmed, and should be revised the moment it is.

Table 9: Provisional demo acceptance gates.

Gate	Provisional threshold
Memory	Model-process peak $\leq$ 1.3 GB; revise after app RAM is confirmed.
Placement	No unsupported main-backbone op; no undeclared CPU fallback in NPU shards.
ASR quality	Quantized WER increase $\leq$ 0.3 pp on each evaluation split.
Responsiveness	First audible PCM $\leq$ 500 ms after VAD close on the actual device.
Context	Two-turn retention no more than five points below the BF16 golden set.
Audio	No NaN/Inf, truncation, or material audio-text consistency regression.

9.2 Next steps

1. Confirm the physical glasses SoC, app-available RAM, QNN/Voice AI SDK version, and deployment interface.
2. Profile Qwen3-0.6B first, then the Moonshine Base encoder and cached decoder, then Pocket TTS on the official host path. Keep Zipformer and Piper ready as swaps.
3. Export four-layer LFM backbone shards and a production cached decoder with explicit position/mask inputs and bounded context.

4. Convert and profile the target-QNN quantized encoder, backbone, depth decoder, and caches separately. Re-run transcript and token checks after every precision change.
5. Measure the official FP32 detokenizer on the intended CPU/DSP path. Do not move it to FP16 HTP until real-code audio fidelity passes.
6. Integrate VAD and measure speech-end to first audible PCM, process RSS, 30-minute temperature, and battery drop on the real glasses.

10 Limitations

The proxy is not the target. QCS8550 latency is not AR1 latency, and no AR1/AR1+/5100 device was available for any measurement in this report.

Component latency and memory cannot be summed into an end-to-end figure. The components have different shapes, invocation counts, compiled-context overheads, and transfer boundaries, and Figure 2 should not be read as a budget.

Backbone evidence is representative rather than one full compiled network. The ASR diagnostic uses 18 utterances and synthetic noise, and is not a glasses microphone benchmark. Host FP32 detokenizer viability is proposed, not measured on target. The Q4 matrix uses two official samples on Apple M2 Metal and establishes neither general quality, nor Qualcomm placement, nor glasses memory.

Finally, licensing is not settled: the LFM Open License v1.0 contains a commercial-use revenue threshold, and the candidate components carry separate attribution and license terms.

References

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A Reproducibility

Table 10: Pinned revisions and environments. Hashes are unbroken for copy-paste.

Item	Pinned value
LFM model revision	c362a0625dfe45aa588dce5f0ada28a7e5707628
Q4 GGUF / runner	7d525f883a077e20afb782f2ff618edcae0e39e4; build 7641 (68d8edf2); Apple M2 Metal
Liquid Audio source	19e65845923a7f136442c95137884ec61eb386aa
Qualcomm proxy	QCS8550 (Proxy), Android 12, Hexagon v73, QNN HTP FP16
Comparator catalog	qai-hub-models 0.57.3; QAIRT 2.45.0 scorecards
Full-model golden	PyTorch 2.8.0, Transformers 4.56.1, NVIDIA L4 BF16